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Studierichting: Informatica
Oefeningenreeks 4A nr. 48.3

$$\begin{aligned} & \int \ln(x^2 - 1) dx \\ & \left\{ \begin{array}{l} u = \ln(x^2 - 1) \\ dv = dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \frac{2x}{x^2 - 1} dx \\ v = x \end{array} \right. \\ & = x \ln(x^2 - 1) - \int \frac{2x^2}{x^2 - 1} dx \\ & = x \ln(x^2 - 1) - 2 \int \frac{x^2 - 1 + 1}{x^2 - 1} dx \\ & = x \ln(x^2 - 1) - 2 \left(\int dx + \int \frac{dx}{x^2 - 1} \right) \\ & = x \ln(x^2 - 1) - 2 \left(x + \int \frac{dx}{x^2 - 1} \right) \end{aligned}$$

Fuss toepassen:

$$\int \frac{dx}{x^2 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1}$$

Voor $x = 1$:

$$A = \frac{1}{1 + 1} = \frac{1}{2}$$

Voor $x = -1$:

$$\begin{aligned} B &= \frac{1}{-1 - 1} = -\frac{1}{2} \\ &= x \ln(x^2 - 1) - 2 \left(x + \frac{1}{2} \int \frac{dx}{x - 1} - \frac{1}{2} \int \frac{dx}{x + 1} \right) \\ &= x \ln(x^2 - 1) - 2 \left(x + \frac{1}{2} \ln|x - 1| - \frac{1}{2} \ln|x + 1| \right) + C \\ &= x \ln(x^2 - 1) - 2x - \ln|x - 1| + \ln|x + 1| + C \\ &= x \ln(x^2 - 1) - 2x + \ln \left| \frac{x + 1}{x - 1} \right| + C \end{aligned}$$

References